ROAD INSPECTOR

ROAD QUALITY INSPECTION
&
REPAIR ESTIMATION SYSTEM

Project ID:-R26-IT-023

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AI & IoT-Based Road Defect Detection System



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AI-Based Repair Area Estimation using Semantic Segmentation



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Automated Material Requirement Estimation for Road Repairs



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Environmental and Traffic Analysis with Repair Scheduling and Alternate Route Recommendation




ROAD INSPECTOR

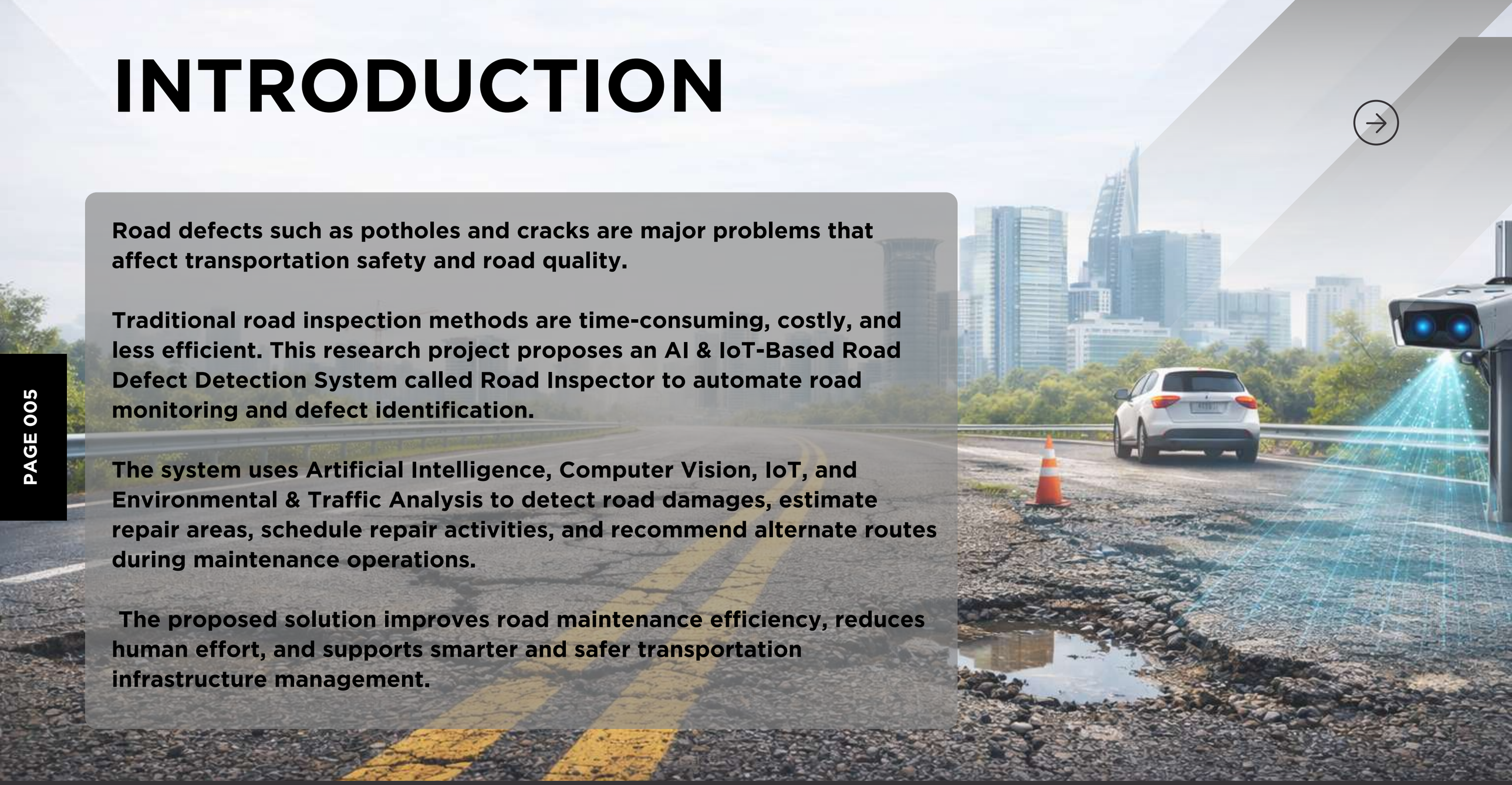
INTRODUCTION

Road defects such as potholes and cracks are major problems that affect transportation safety and road quality.

Traditional road inspection methods are time-consuming, costly, and less efficient. This research project proposes an AI & IoT-Based Road Defect Detection System called Road Inspector to automate road monitoring and defect identification.

The system uses Artificial Intelligence, Computer Vision, IoT, and Environmental & Traffic Analysis to detect road damages, estimate repair areas, schedule repair activities, and recommend alternate routes during maintenance operations.

The proposed solution improves road maintenance efficiency, reduces human effort, and supports smarter and safer transportation infrastructure management.





USERS AND STAKEHOLDERS

Road Maintenance Planners

- Plan repair operations.

Municipal Road Authorities

- Monitor road conditions and manage maintenance budgets.

Maintenance Field Teams

- Perform road repair activities.

Material Estimation Engineers

- Calculate required construction materials.

Road Users

- Drivers and pedestrians affected by unsafe roads

IMPACT OF THE PROBLEM



1. SOCIAL IMPACT

- Increased risk of accidents
- Unsafe travel conditions

3. TRANSPORTATION IMPACT

- Traffic congestion
- Travel delays

2. ECONOMIC IMPACT

- Vehicle damage
- Higher maintenance costs

4. ENVIRONMENTAL IMPACT

- Waste of construction materials
- Increased emissions due to inefficient repairs

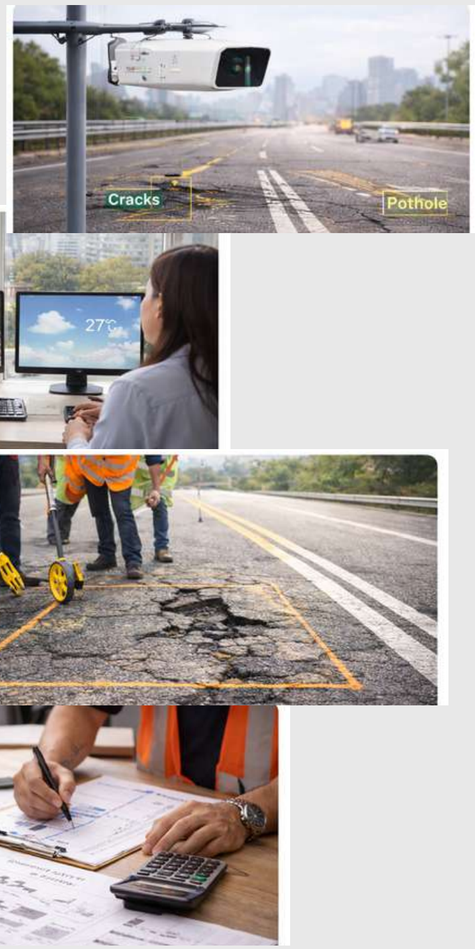
PROPOSED SOLUTION

AI-Driven Road Inspection and Decision Support System

We propose an AI-powered platform that provides:

- Automated road defect detection using computer vision
- Defect severity assessment
- Accident risk prediction
- Repair geometry estimation
- Material requirement estimation
- Intelligent repair scheduling

The system integrates AI, IoT data collection, and machine learning to transform traditional road inspection into a smart maintenance process.



RESERCH INTRODUCTION & NOVELTY

Problem

• Current road inspection processes mainly depend on manual visual assessment, causing inconsistent defect identification and severity evaluation.
• Existing AI-based road damage detection systems mainly focus on defect detection only, without providing complete maintenance decision support.
• Lack of integration between defect detection, risk assessment, repair estimation, and maintenance planning creates delays and inefficient resource allocation.

Our Research

AI & IoT-Based Smart Road Maintenance Decision Support System

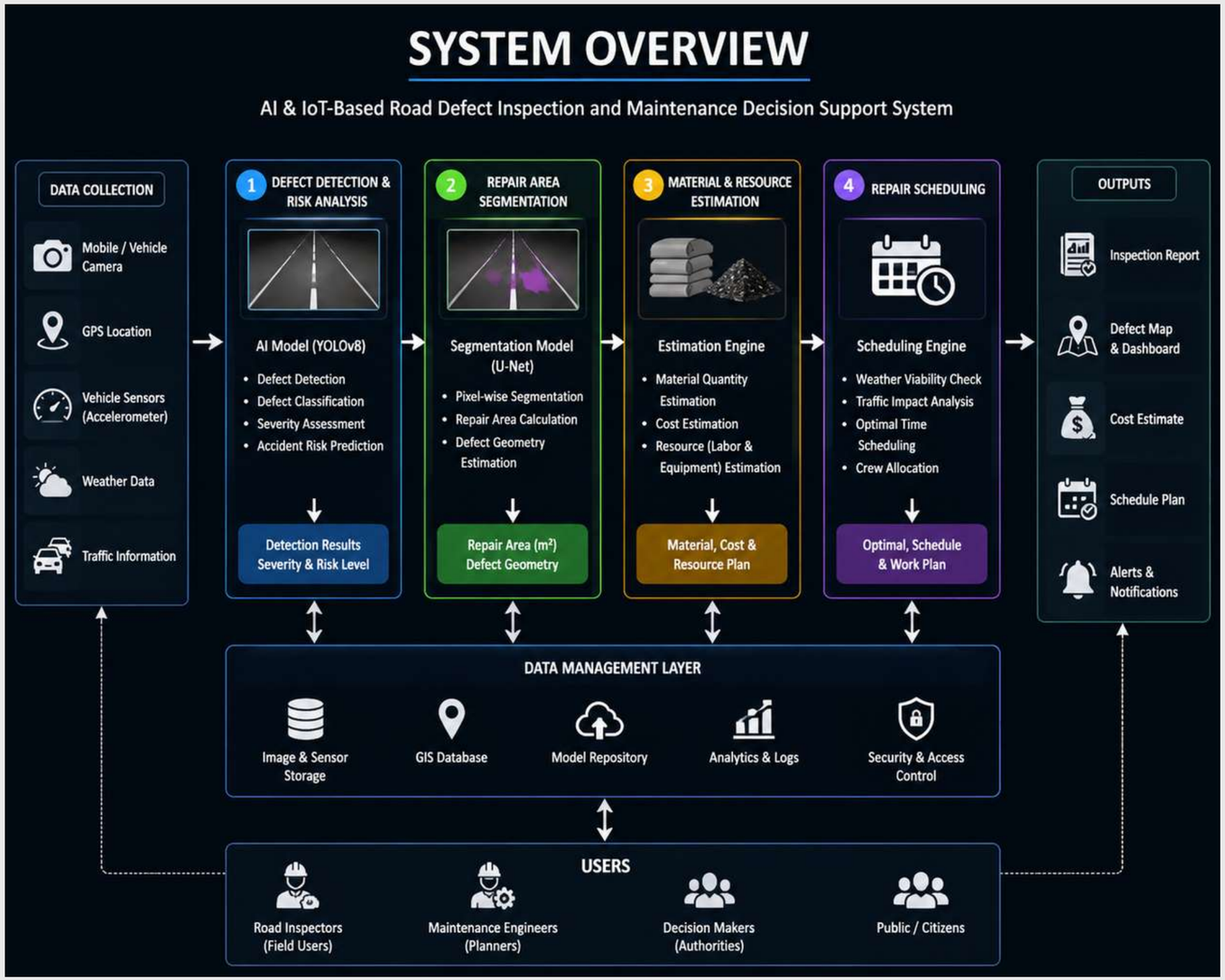
• AI-Based Road Condition Monitoring and Defect Analysis
• Intelligent Assessment of Road Damage Severity and Risk Levels
• Automated Repair Planning and Resource Optimization
• Real-Time Data Integration for Smart Maintenance Decisions
• End-to-End AI-Driven Road Inspection and Management Support



Novelty

A unified AI-driven road inspection and maintenance decision support system that integrates defect detection, severity analysis, accident risk prediction, repair estimation, and scheduling into a single platform.
Unlike existing approaches that only identify road defects, this research focuses on transforming visual road data into actionable maintenance decisions with risk-aware recommendations.

OVEROLL ARCHITECTURE & RESERCH FLOW



FOUR COMPONENTS & CONTRIBUTION

1. AI-Based Road Defect Detection & Accident Risk Analysis → Detects road defects, classifies damage severity, and predicts accident risks using AI-based computer vision and contextual data.

2. AI-Based Repair Area Estimation Using Semantic Segmentation → Generates pixel-level defect masks to accurately estimate damaged areas and repair geometry for maintenance planning.

3. Automated Material Requirement Estimation for Road Repairs → Calculates required repair materials, resources, and cost based on defect measurements and repair requirements.

4. Environmental & Traffic Analysis with Repair Scheduling and Alternate Route Recommendation → Analyzes weather and traffic conditions to optimize repair timing and recommend alternative routes to minimize disruption.

Component Novelty

- **AI-Based Defect Detection:** Automated identification, classification, severity assessment, and accident risk analysis of road defects.
- **Semantic Segmentation-Based Repair Estimation**: Converts detected defects into measurable repair areas instead of simple visual detection.
- **Intelligent Resource Estimation:** Provides automated material, labour, and cost estimation to support maintenance decisions.
- **Smart Scheduling & Route Optimization:** Enables proactive repair planning by integrating environmental and traffic intelligence.

Functional & Non-Functional Requirements



Functional Requirements

- **Data Collection:** Collect and process required data from different sources.
- **Data Processing:** Analyze and transform collected data into meaningful information.
- **Intelligent Analysis:** Apply AI/ML techniques to identify patterns, detect issues, and generate predictions.
- **Decision Support:** Provide recommendations, insights, and automated assistance to users.
- **System Interaction:** Allow users to access system features through an interactive interface.
- **Reporting & Alerts:** Generate reports, notifications, and alerts based on system outputs.

Non-Functional Requirements

- **Accuracy:** Provide reliable and precise results.
- **Performance:** Deliver results with efficient response time.
- **Scalability:** Support increasing data volume and user requirements.
- **Security:** Protect system data and user information.
- **Reliability:** Ensure stable and continuous system operation.
- **Usability:** Provide a simple and user-friendly interface.
- **Maintainability:** Support future updates and system improvements.





VALIDATION & EVALUATION

- **Model Evaluation:** Assess the accuracy and effectiveness of developed models and algorithms.
- **Performance Measurement:** Evaluate system performance using suitable evaluation metrics.
- **Data Validation:** Test the system using unseen data to ensure reliability and generalization.
- **System Testing:** Verify the functionality, integration, and overall performance of the complete system.
- **Real-World Testing:** Validate the system using practical scenarios and real-world conditions.
- **User Evaluation:** Assess usability, user satisfaction, and practical usefulness of the system.



ABILITY OF COMMERCIALIZATION

Target Market:
Road development authorities, municipal councils, highway maintenance departments, and smart city infrastructure projects.

Value:
AI-driven road defect detection, repair area estimation, material estimation, and smart maintenance planning to improve road maintenance efficiency and reduce operational costs.

Revenue Model:
Subscription-based platform services, government infrastructure solutions, maintenance analytics services, and smart road monitoring solutions.

Advantage:
Integrated AI, IoT, semantic segmentation, traffic analysis, and material estimation in a single intelligent road maintenance platform.

Market Potential:
Supports the growing demand for smart transportation infrastructure and automated road maintenance systems in modern cities and developing countries.


THANK YOU.